

Operations Research

Linear Programming Models

Simplex method

In simplex method, the problem is always maximization type

the constants are " $\leq$ ".

Eg.1: Maximize $Z = x_1 + 4x_2$ subject to

$$2x_1 + x_2 \leq 3$$

$$3x_1 + 5x_2 \leq 9$$

$$x_1 + 3x_2 \leq 5 \quad \text{and } x_1, x_2 \geq 0$$

Objective function Max. $Z = x_1 + 4x_2$ s.t

Constraints

$$2x_1 + x_2 \leq 3$$

$$3x_1 + 5x_2 \leq 9$$

$$x_1 + 3x_2 \leq 5$$

There are three constraints, we will introduce three slack variables in the above equation, (s_1, s_2, s_3)

Step 1: Standard form of equation is

$$\text{Max. } Z = x_1 + 4x_2 - 0s_1 - 0s_2 - 0s_3$$

Taking the variables in the objective function to the left side, we get

$$\text{Max. } Z - x_1 - 4x_2 + 0s_1 + 0s_2 + 0s_3 = 0$$

$$2x_1 + x_2 + 1s_1 + 0s_2 + 0s_3 = 3$$

$$3x_1 + 5x_2 + 0s_1 + 1s_2 + 0s_3 = 9$$

$$x_1 + 3x_2 + 0s_1 + 0s_2 + 1s_3 = 5$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Table 1

	Basic variable	Coefficients A					RHS	RHS key Column
		x_1	x_2	s_1	s_2	s_3		
R_1	Z	-1	-4	0	0	0	0	$\frac{0}{-4} = 0$
R_2	s_1	2	1	1	0	0	3	$\frac{3}{1} = 3$
R_3	s_2	3	5	0	1	0	9	$\frac{9}{5} = 1.8$
R_4	s_3	1	3	0	0	1	5	$\frac{5}{3} = 1.67$

Consider coefficients on the top row (Table 1)

Identify the coefficient that is more negative

-4 is more negative than -1

Circle the column containing the more -ve number,

-4. This column is called key column

Divide the RHS by key column, we get

$$\frac{0}{-4} = 0$$

$$\frac{3}{1} = 3$$

$$\frac{9}{5} = 1.8$$

$$\frac{5}{3} = 1.67$$

of these, the smallest positive ratio is 1.67

We select this row

The row containing smallest positive ratio is called key row. Here the smallest positive ratio is $\frac{5}{3}$

The intersection of key row and key column is called key element

In this problem, the key element is 3.

Now, we have to make key element 1. This is possible by dividing the entire row by 3.

Table 2:

	Basic variable	Coefficient of					RHS
		x_1	x_2	s_1	s_2	s_3	
R_1	z	-1	-4	0	0	0	0
R_2	s_1	2	1	1	0	0	3
R_3	s_2	3	5	0	1	0	9
R_4	s_3	$\frac{1}{3}$	1	0	0	$\frac{1}{3}$	$\frac{5}{3}$

With the help of key element, convert the rows above and below key element to zero.

With the help of R_4 , make R_3 zero

$$R_3 \rightarrow R_3 - 5R_4 \quad 3 - 5\left(\frac{1}{3}\right) = 3 - \frac{5}{3} = \frac{9-5}{3} = \frac{4}{3}$$

$$5 - 5(1) = 5 - 5 = 0 \quad 0 - 5(0) = 0 - 0 = 0$$

$$1 - 5(0) = 1 - 0 = 1$$

$$0 - 5\left(\frac{1}{3}\right) = -\frac{5}{3}$$

$$9 - 5\left(\frac{5}{3}\right) = 9 - \frac{25}{3} = \frac{27-25}{3} = \frac{2}{3}$$

With the help of R_4 , make R_2 zero.

$$R_2 \rightarrow R_2 - R_4 \quad 2 - \frac{1}{3} = \frac{6-1}{3} = \frac{5}{3}$$

$$1 - 1 = 0$$

$$1 - 0 = 1$$

$$0 - 0 = 0$$

$$1 - \frac{1}{3} = -\frac{1}{3}$$

$$3 - \frac{5}{3} = \frac{9-5}{3} = \frac{4}{3}$$

With the help of R_4 , make R_1 zero

$$R_1 \rightarrow R_1 + 4R_4$$

$$-4 + 4(1) = -4 + 4 = 0$$

$$-1 + 4\left(\frac{1}{3}\right) = -1 + \frac{4}{3} = \frac{4}{3} - 1 = \frac{1}{3}$$

$$0 + 4(0) = 0 + 0 = 0$$

$$0 + 4(0) = 0 + 0 = 0$$

$$0 + 4\left(\frac{1}{3}\right) = \frac{4}{3}$$

$$0 + 4\left(\frac{5}{3}\right) = \frac{20}{3}$$

Table 3 :

	Basic variable	Coefficient of					RHS
		x_1	x_2	s_1	s_2	s_3	
R_1	z	$\frac{1}{3}$	0	0	0	$\frac{4}{3}$	$\frac{20}{3}$
R_2	s_1	$\frac{5}{3}$	0	1	0	$-\frac{1}{3}$	$\frac{4}{3}$
R_3	s_2	$\frac{4}{3}$	0	0	1	$-\frac{5}{3}$	$\frac{2}{3}$
R_4	s_3	$\frac{1}{3}$	(1)*	0	0	$\frac{1}{3}$	$\frac{5}{3}$

←
out
 x_2
replace
 s_3

There is no negative term in Row 1 (R_1)

$$\text{we have value } z = \frac{20}{3}$$

In Table 2, consider key element 3, s_3 is the at the key row and x_2 is at the ^{key} column. (5)

s_3 will leave and x_2 will enter replacing s_3

	Base Variable	Coefficients of					RHS
R_1	Z	x_1	x_2	s_1	s_2	s_3	
		$1/3$	0	0	0	$4/3$	$20/3$
R_2	s_1	$5/3$	0	1	0	$-1/3$	$4/3$
R_3	s_2	$4/3$	0	0	1	$-5/3$	$2/3$
R_4	x_2	$1/3$	1	0	0	$1/3$	$5/3$

$s_3 \leftarrow$ From the above $x_2 = 5/3$ $\text{Max } Z = 20/3$

$$x_1 = ?$$

$$\text{Max } Z = x_1 + 4x_2$$

$$\frac{20}{3} = x_1 + 4\left(\frac{5}{3}\right)$$

$$\frac{20}{3} = x_1 + \frac{20}{3}$$

$$x_1 = \frac{20}{3} - \frac{20}{3} = 0$$

$$\therefore x_1 = 0$$

The solution is $Z_{\text{max}} = \frac{20}{3}$, $x_1 = 0$, $x_2 = \frac{5}{3}$